AI-POWERED ONCOLOGY

Deep Neural Networks for Classifying Malignant & Benign Breast Tumors

CONTEXT & CLINICAL OBJECTIVES

Early detection dramatically transforms breast cancer survival rates. Automated deep learning classification minimizes diagnosis latency and pathologist bottleneck.

THE CLINICAL PARADIGM

Pathology Bottleneck

Fine Needle Aspirate (FNA) diagnostic evaluations rely on manual cytological evaluation under standard microscopes. This introduces critical visual subjectivity, leading to potential variance in tumor classification.

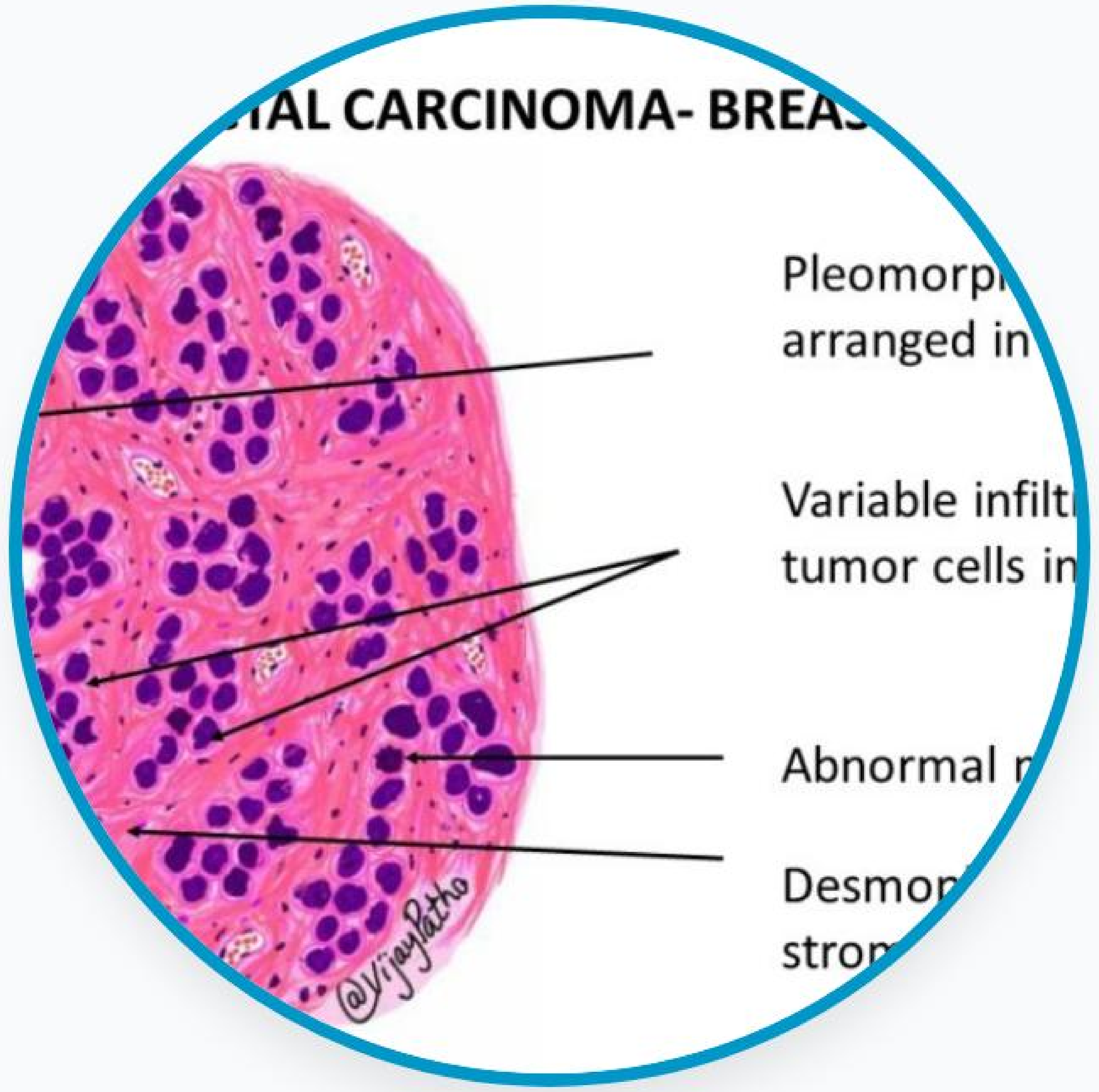
Consequences of a False Negative error are severe, causing critical delays in initiating life-saving targeted treatments.

Deep Learning Support

Multi-layered artificial neural networks compute localized non-linear dependencies across nuclear morphology to classify malignant tumors automatically.

The network functions as a reliable, instant diagnostic co-pilot, enhancing overall cytological accuracy and streamlining specialist workflow.

FINE NEEDLE ASPIRATES



Wisconsin Cytology Dataset

Our model leverages the benchmark Wisconsin Diagnostic Breast Cancer (WDBC) dataset. This dataset features 30 highly descriptive morphometric attributes extracted from digitised aspirate images.

Key biological properties captured include nuclear radius, cell boundary perimeter, area distribution, perimeter texture, local cell smoothness, compactness, and concave nuclear markers.

END-TO-END PIPELINE

Data Ingestion

FNA cytological features are mapped into 30 numerical dimensional vectors.

Standardization

Z-score scaling centers feature values to prevent dominant magnitude variance.

Model Learning

An 80/20 train-test partition updates weights via Binary Cross-Entropy minimization.

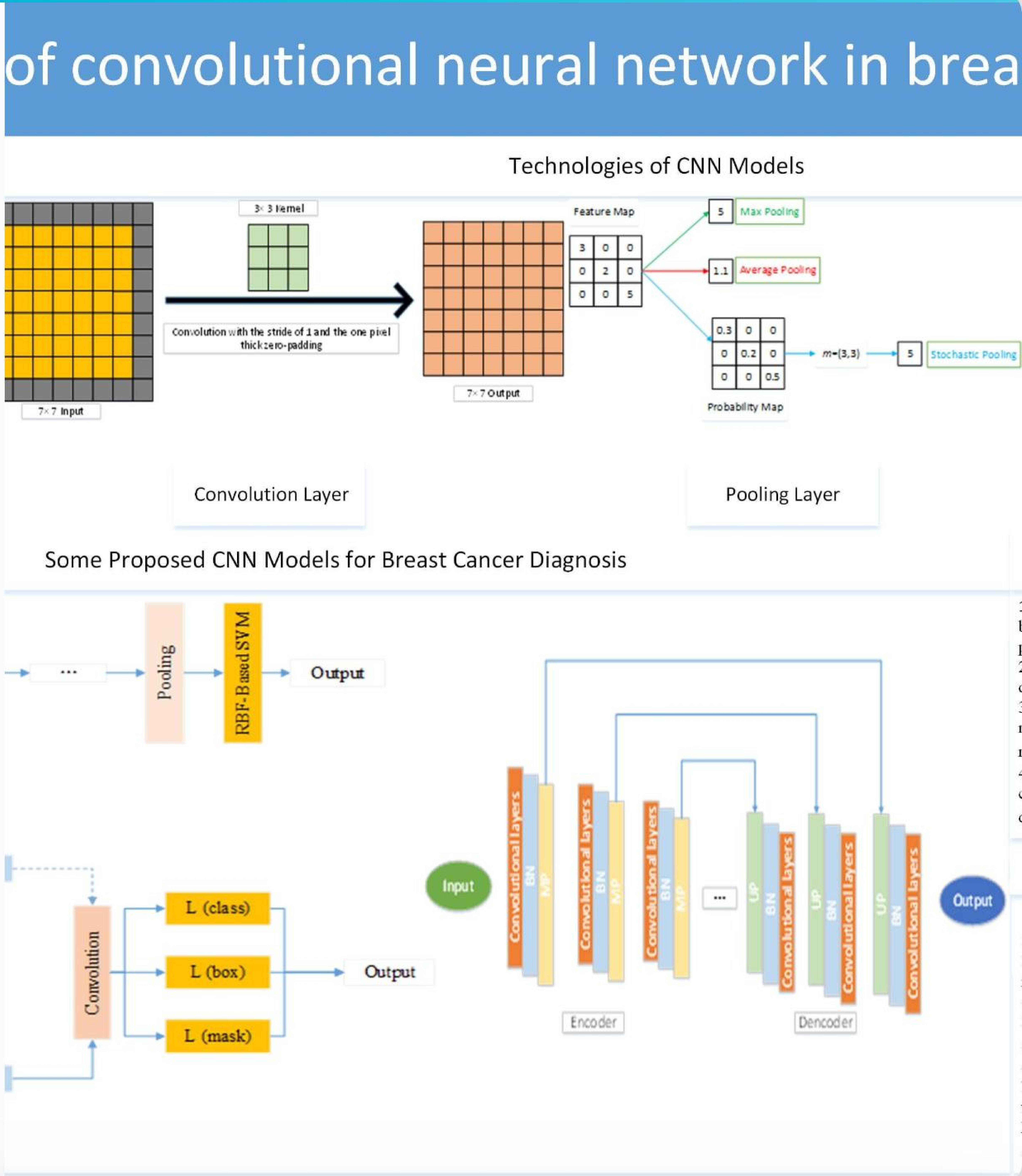
Evaluation

Performance metrics are calculated, prioritizing clinical sensitivity.

FEATURE EXTRACTION

Multi-Layered Abstraction

Artificial neural networks automatically map non-linear relationships across high-dimensional cellular features. Unlike traditional manual classification, a multi-layer perceptron constructs robust decision boundaries to segment benign from malignant cellular clusters with maximum precision.



NETWORK ARCHITECTURE

Input Layer

Consists of 30 neurons mapping raw dimensional morphology measurements (radius, texture, concavity) directly from FNA cells.

Hidden Layers

Two fully-connected Dense Layers (32 and 16 units) utilizing Rectified Linear Unit (ReLU) activation to capture complex cellular shapes.

Output Layer

A single Sigmoid node computing real-time malignancy probabilities.

$$f(z) = \frac{1}{1+e^{-z}}$$

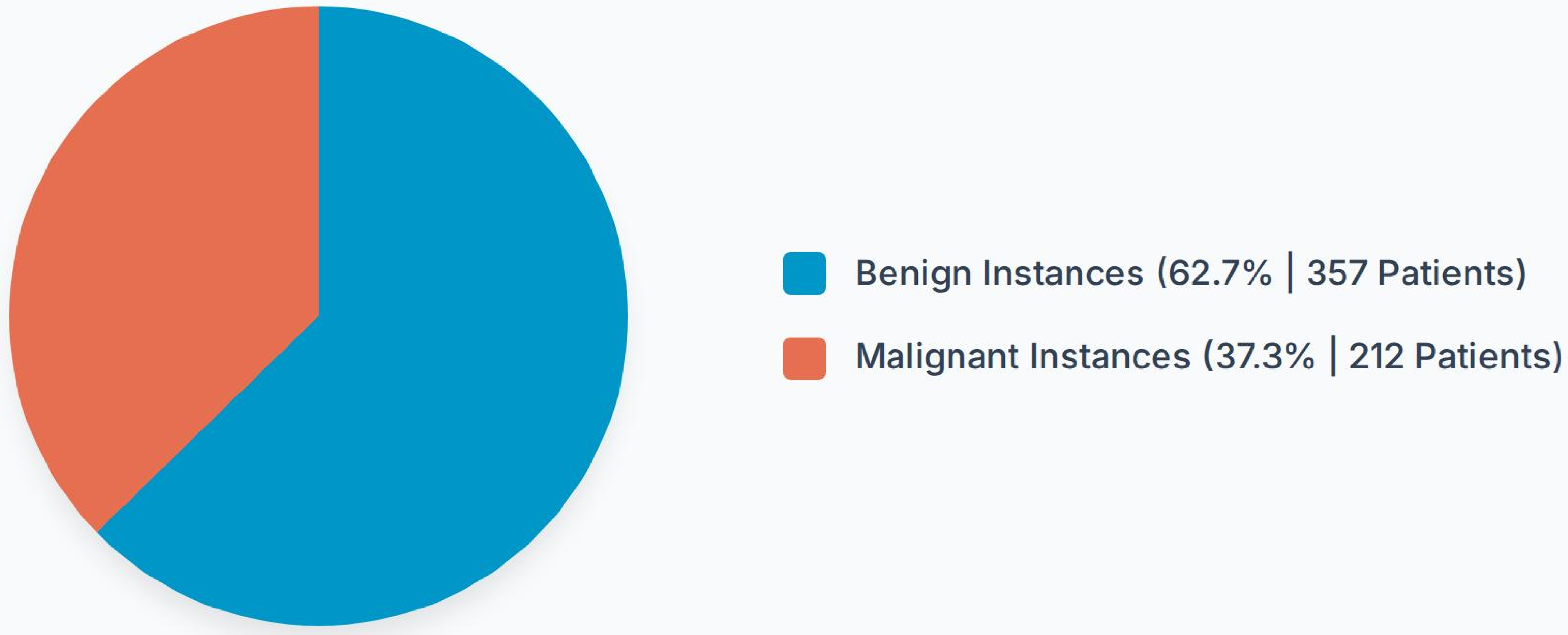
HYPERPARAMETER CONFIGURATIONS

Hyperparameter	Standard Setup	Optimized Network	Clinical Target
Optimizer Engine	SGD (Stochastic Gradient Descent)	Adam (Adaptive Moment Estimation)	Fast error-surface convergence
Loss Evaluation Function	Mean Squared Error	Binary Cross-Entropy	Maximizes classification margin
Hidden Activation	Sigmoid Layer	ReLU (Rectified Linear Unit)	Mitigates vanishing gradients
Learning Rate Decay	0.01 (Constant)	0.001 (Adaptive schedule)	Stabilizes loss minimisation
Epoch Training Limit	10 Cycles	50 Cycles	Avoids model overfitting

CLASSIFICATION PERFORMANCE

The neural network (ANN) demonstrates superior generalization across complex non-linear cytological boundaries, outperforming standard machine learning classifiers by establishing wider discriminative margins.

PATIENT CLASS BALANCE



A balanced representation allows the neural network to avoid majority-class bias, ensuring the sensitivity rate remains clinically viable for identifying critical malignant clusters.

CLINICAL TRANSLATION

> "Deep learning models do not aim to replace clinical pathologists. Instead, they act as high-precision assistants to filter out true negatives and flag complex malignancies, reducing diagnostic turn-around times."

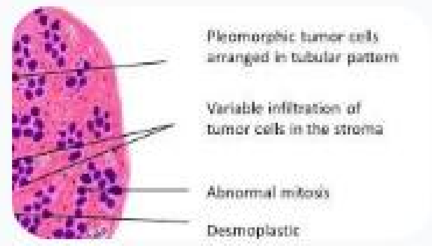
— Lead Research Clinician, Oncological AI Lab

QUESTIONS & ANSWERS

Enhancing Diagnostic Certainty with Neural Networks

Oncology Informatics Department | ml-oncology@clinical-research.org

IMAGE SOURCES



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